

Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.52804	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	77.63892	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1447.12264	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.273435	0.273435	136.76162	0.0	0.0

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.52804.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.

- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 77.63892.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1447.12264.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 136.76162.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]

- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Phase-9 CVRP Solver-Evolution Suite Summary

Metric	Baseline Nearest Neighbor	Baseline Clarke-Wright Savings	Baseline Regret Insertion LS	Evolved Solver (Phase 9)
Feasibility Rate	100%	100%	100%	100%
Heldout Feasible Gap	0.2734	**0.0738 (best)**	0.4664	0.2734
Heldout Penalized Gap	0.2734	0.0738	0.4664	0.2734
Heldout Runtime (ms)	42.5	77.6	1447.1	136.8
Robustness Dispersion	0.0028	0.0078	0.0869	0.0028

- ****Feasibility****: All methods, including the evolved solver, achieved perfect feasibility (100%) on held-out instances.
- ****Objective Gap****: The evolved solver's gap (0.2734) matches the baseline nearest neighbor constructive method but is significantly worse than the Clarke-Wright baseline (0.0738), which had the best solution quality.
- ****Runtime****: The evolved solver (137 ms) is slower than nearest neighbor (43 ms) and Clarke-Wright (78 ms) but much faster than regret insertion local search (1447 ms).
- ****Robustness****: Evolved solver's robustness in held-out performance dispersion is low (0.0028), similar to nearest neighbor and lower than Clarke-Wright and regret insertion.

Conclusion

The evolved solver ****did not clearly beat the fixed baselines****. It matched nearest neighbor in feasibility, gap, and robustness but was outperformed by Clarke-Wright Savings in terms of objective gap and by nearest neighbor in runtime. It was substantially faster and more robust than regret insertion local search but with worse gap quality. Thus, while robust and feasible, the solver evolution did not produce a clearly superior solver over the best fixed baseline (Clarke-Wright Savings).